GAURAV DHAK

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PROFESSIONAL SUMMARY

RTL and FPGA Engineer with experience in Verilog-based design, FPGA prototyping, and VLSI design flow. Developed hardware accelerators, cryptographic cores, and digital design projects with focus on verification, timing closure, resource optimization, and AXI4-based system integration.

TECHNICAL SKILLS

- **Programming:** C/C++, Python, Linux Shell Scripting
- **Hardware Description Languages:** Verilog
- **EDA Tools:** Xilinx Vivado, Vitis HLS, Cadence Virtuoso, Icarus Verilog
- **FPGA Platforms:** Xilinx Zynq UltraScale+ MPSoC (ZCU104), Arty A7, Basys 3
- **Design & Verification:** RTL Design, RTL Simulation, FPGA Prototyping, Hardware-Software Co-Design, AXI4 Protocol, Timing Closure
- **VLSI:** CMOS Digital Design, VLSI Design Flow

EXPERIENCE

Project Associate — National Institute of Technology, Goa **Nov 2025 – Present**

- Architected and implemented a high-throughput DNN accelerator on FPGA using parallel dataflow and pipelined execution to improve computational throughput and reduce latency.
- Developed and integrated FPGA accelerator modules using Xilinx toolchains, supporting system-level prototyping and validation.
- Performed design-space exploration and analyzed throughput, latency, and FPGA resource utilization across multiple architecture configurations.

PROJECTS

AES-128 Encryption Core Design and FPGA Implementation

- Designed and implemented a fully synthesizable AES-128 encryption core in Verilog, including SubBytes, ShiftRows, MixColumns, and Key Expansion modules.
- Optimized the design for FPGA deployment on Zynq ZCU104, achieving timing closure and efficient resource utilization
- Integrated IV-based encryption mode to improve security against pattern-based attacks.

Verilog-Based Branch Predictor Using 2-Bit Saturating Counters

- Designed and implemented a 2-bit saturating counter-based branch predictor to improve control-flow prediction in a simulated processor pipeline.
- Achieved ~85% prediction accuracy across 100+ dynamic branch patterns and verified functionality using a cycle-accurate testbench in Vivado.

2-Way Set-Associative Cache Memory Simulator

- Developed a 2-way set-associative cache model implementing tag comparison, hit/miss detection, and LRU replacement policy.
- Evaluated cache performance and memory access behavior using a dual-clock (1 MHz) architecture.
- Verified functionality across multiple memory access patterns and cache replacement scenarios.

EDUCATION

Bachelor of Technology in Electronics System Engineering

National Institute of Electronics and Information Technology (NIELIT), Aurangabad | May 2025 | CGPA: 7.82/10

CERTIFICATIONS & ACHIEVEMENT

- Samsung Fellowship – Cohort 4 (ISWDP), IISc Bangalore & Samsung Semiconductor India Research (2025)
- Certified Chip Design Associate (RTL & FPGA) — NIELIT Calicut (2025)
- VLSI Design Flow: RTL to GDS — IIIT Delhi (2024)
- VLSI for Beginners — NIELIT Calicut (2024)